

Generation Interconnection System Impact Study Report

For Queue Project Q21-1726

Alderwood Manor 100 kV

August 25, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1726. The Interconnection Customer proposes a 157.1 MW wind facility interconnecting at the Alderwood Manor 100 kV substation in the Illinois area. The request was received on April 22, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 530 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$100,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Alder-Shady Oaks Wind Farm 230 KV

Contingency: Shady Oaks Wind Farm-Alder Substation 500 KV

Loading: 102.5% (Rating: 1,169 MVA)

DFAX: 6.3%

Pre-project loading: 101.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Alder-Shady Oaks Wind Farm 230 KV	Reconductor line	\$4,478,000	2% (\$100,000)
	Total allocated to project		\$100,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.